

Estimating probabilities of malaria importation in southern Mozambique through *P. falciparum* genomics and mobility patterns

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This study introduces a **useful** method to estimate the probability that a malaria case is imported and to identify the geographic origin of parasites by using a Bayesian approach that integrates epidemiological, travel, and genetic data. The authors provide **convincing** evidence that the approach can reliably identify the main sources of malaria imports. This work will be of great interest to the area of genomic epidemiology and public health strategies aiming to eliminate malaria.

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Abstract

Imported malaria is a critical obstacle to achieving elimination in low transmission settings. Characterising malaria importation and transmission sources using human mobility and parasite genomics has the potential to inform elimination strategies, but tools combining both types of data are lacking. We developed a novel Bayesian approach that provides individual importation probabilities and geographic origin of *P. falciparum* cases by combining epidemiological, human mobility and parasite genetic data. Spatial genetic structure and connectivity were assessed using microhaplotype-based genetic relatedness (identity-by-descent) from 1605 *P. falciparum* samples collected from 9 provinces in Mozambique during 2022, including two very-low transmission elimination-targeted districts (Magude and Matutuine) in the south. Travel reports were combined with genetic relatedness metrics to

classify clinical cases as local or imported. Genetic relatedness between parasites from southern and northern/central Mozambique was lower (0.021) than average (0.034, $p < 0.001$). 42% (88/207) of infections in elimination-targeted districts were classified as imported, had a higher genetic complexity (OR=1.3) and originated mainly from Inhambane (63% [55/88]). Significant differences in importation rates were found between the two studied districts (OR=6.6), with Magude district (10.71%, 3/28) showing lower importation rates than Matutuine (48.60%, 87/179) district. Differences in importation rates observed between both elimination districts suggest the need for fine-scale analysis to tailor cost-effective elimination strategies. Importation is playing a crucial role in sustaining transmission in Matutuine district, and increasing efforts to reduce malaria burden in their sources of transmission (especially in Inhambane province), as well as targeting travelers to central and northern Mozambique, could significantly contribute to malaria elimination in the south.

Introduction

Despite the efforts during the last decades, malaria elimination remains challenging, especially in Sub-Saharan Africa^{1–6}. Detection, characterisation and monitoring of malaria infections are key to control and elimination⁷. However, decisions become more complex in low transmission areas given the heterogeneity in transmission patterns, requiring better tools to identify the key sources of transmission⁸.

One of the key challenges for malaria elimination in very-low transmission settings is parasite importation, which can sustain transmission during “last mile” efforts^{9–14}. Identifying imported cases and their transmission sources, together with associated risk factors, is key to improving targeted efforts for malaria elimination in very-low transmission areas. Previous studies that aimed to classify reported cases as local or imported used epidemiological data based on travel reports (assuming or modelling that the infections occurred during the trip)^{15–18}, or mobile phone data and geospatial modelling to characterise population mobility and infer the potential impact of importation^{19–22}. Other studies used parasite genomics to assess the spatial connectivity of genetic populations to infer transmission flow, migration patterns or infection origin^{23–37}. However, a study comparing mobility, phone and parasite genetic data brought distinct conclusions on the spatial connectivities due to intrinsic biases of the different data sources³⁰. Two studies which combined mobility (travel reports and mobile phone data) with genetic data found a positive association between mobility and parasite genetic relatedness, providing evidence of importation^{21,30}. However, none of these studies combined mobility and genetic data simultaneously to provide importation rates in the populations or for individual case classification.

We present a new method to provide case classification as individual importation probabilities by combining travel, epidemiological, and *P. falciparum* genetic data. The study uses data from 9 (out of 11) provinces of Mozambique to assess the spatial structure (dependence on pairwise geographical distance) and differentiation (across areas) of *P. falciparum* genetic populations in the country, quantify the levels of importation in the very low transmission districts of Magude and Matutuine in southern Mozambique, and identify sources of transmission and risk factors associated with human mobility and malaria importation.

Results

Participant recruitment, sample and data collection

The study was conducted during 2022 in 7 health facilities (HF) of Magude and 13 HF of Matutuine districts, both in Maputo province. All the individuals attending the HF with signs and symptoms of malaria and a positive malaria rapid diagnostic test were included in the study if residing in the study area (Suppl. Fig. 1). These are very-low transmission areas (with less than 52 yearly cases per 1,000 people in all HFs) with the highest rainfall between January and May. A total of 809 *P. falciparum* positive clinical cases were reported, from which 609 (75.3%, 609/809) rapid diagnostic tests (RDTs) were available for parasite sequencing and 540 (66.7%, 540/809) were sequenced successfully (with allele calls passing both negative controls and allele frequency filters, see Methods). Demographic data and travel reports were available for 232 (28.7%, 232/809) of the samples, and 207 (25.6%, 207/809) resided in the area and passed sequencing coverage and depth requirements (at least 50 loci with >100 reads, see Methods section for more details)^{38,39} (Table 1, Fig. 1). From these 207, 51.7% (107/207) reported a trip during the last 28 days, with 55.1% (59/107) of those having travelled to Inhambane province, the main travel destination (Fig. 2, Suppl. Table 1). The other principal destinations were Zambézia (14.0%, 15/107), Gaza (12.1%, 13/107) and Maputo provinces (9.3%, 10/107). Significant differences were found between clinical cases from Magude and Matutuine districts on travelling rates ($p < 0.001$), with those from Matutuine showing higher travel rates (10.7% [3/28] in Magude versus 58.1% [104/179] in Matutuine, $p < 0.001$), and occupation ($p = 0.001$), but not on season ($p = 0.060$), age ($p = 0.599$), sex ($p = 0.272$), pregnancy ($p = 1.000$) or travel destination ($p = 0.357$) (Table 1).

Additionally, 1,065 dried blood spot samples from *P. falciparum* uncomplicated clinical malaria cases were collected in 9 provinces during the 2022 rainy season, in the context of annual health facility surveys (HFS) or clinical trials⁴⁰ (Suppl. Figure 1, Suppl. Table 2). The number of samples sequenced per province ranged between 78 and 100, with the exception of Inhambane province which included 364 samples for a deeper spatial analysis in this province (Suppl. Table 2). No significant differences between provinces were found in terms of age ($p = 0.551$) or gender ($p = 0.575$). Analysis were conducted both at province and regional level, where provinces were classified as follows: south (Maputo, Inhambane), centre (Sofala, Manica, Tete, Zambézia) and north (Nampula, Niassa, Cabo Delgado).

Country-wide spatial trends of *P. falciparum* genetic relatedness

DNA extracted from DBS or RDTs was sequenced using the MAD⁴HaTeR targeted amplicon sequencing panel and 165 microhaplotype loci were used to calculate diversity metrics⁴¹. A significant spatial pattern of genetic relatedness (R , defined as the fraction of related (identity-by-descent, IBD) pairs ($IBD > 0.1$ and $p < 0.05$), see Methods) in *P. falciparum* populations within and across provinces was found, with a strong south-centre/north differentiation (Fig. 3). Sample pairs within centre/north Mozambique presented a higher R (0.06, 10238/181503 pairs) than the average (0.03, 41413/1287210 pairs; $p < 0.001$). However, R between south and centre/north sample pairs was lower than average (0.02, 12462/604206 pairs, $p < 0.001$) (Suppl. Fig. 2). The highest R was found within Nampula ($R = 0.12$ [475/4005 pairs]), and the highest R from Maputo province was with Inhambane ($R = 0.04$ [8855/231410 pairs]). When stratifying Maputo and Inhambane provinces by district (samples from these provinces were collected in two districts), R was higher for sample pairs between Matutuine and Inhambane ($R > 0.04$ regardless of the district from Inhambane) than for sample pairs within Matutuine (0.03, 508/15400 pairs) ($p < 0.001$) (Suppl. Fig. 3). No significant differences in R were found comparing parasite samples from Magude and the rest of the districts.

A strong spatial correlation of R at the inter-province scale was found (Suppl. Fig. 4A), with R significantly decreasing with the pairwise geographical distance for distances larger than 100 km ($p < 0.001$). However, no significant correlation was found for distances between 10 km and 100 km ($p = 0.547$, $p = 0.628$ and $p = 0.794$ when using IBD thresholds of 0.1, 0.15 and 0.2 respectively, showing

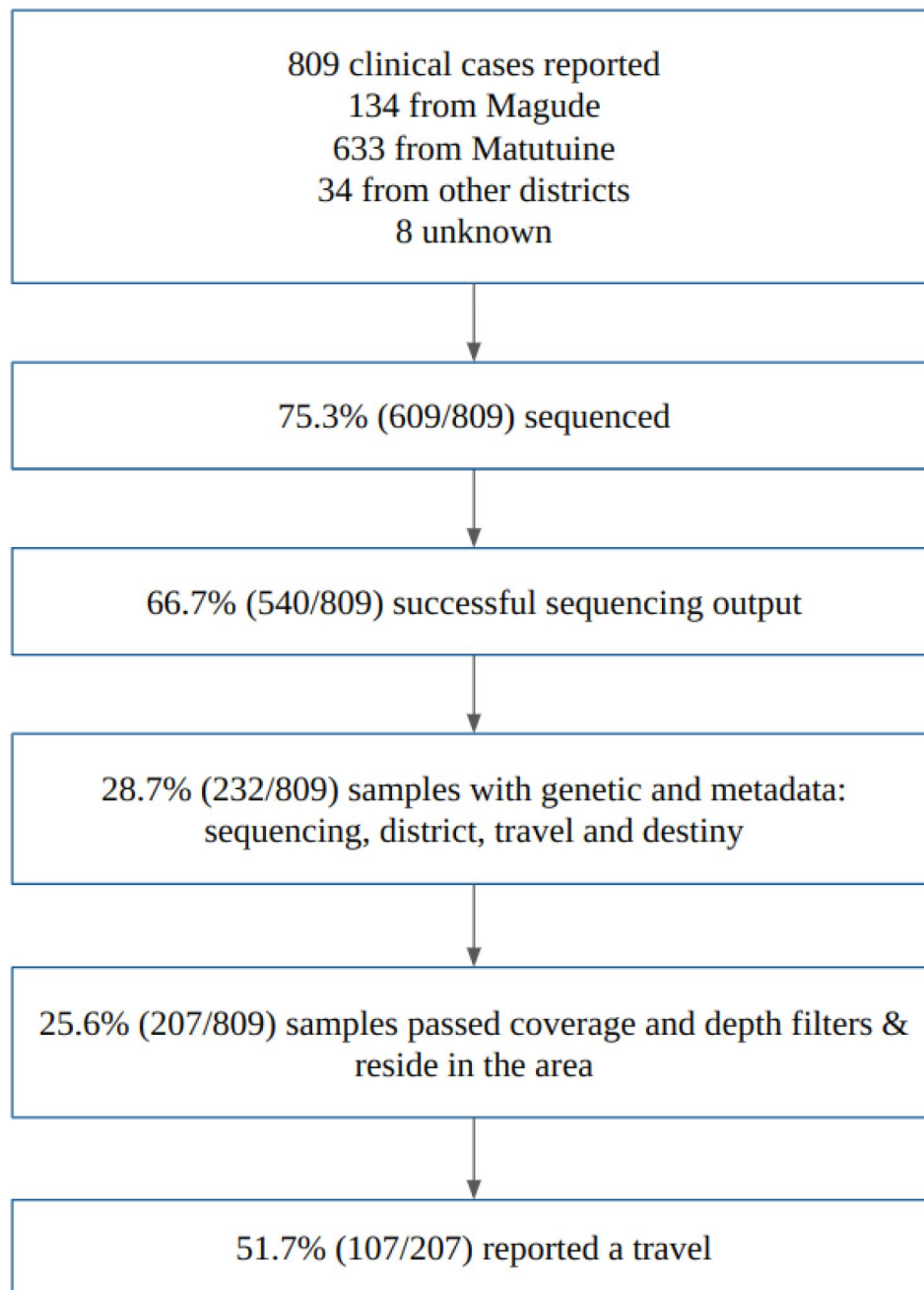


Figure 1

Flowchart of the *P. falciparum* samples and data from Magude and Matutuine districts collected in 2022.

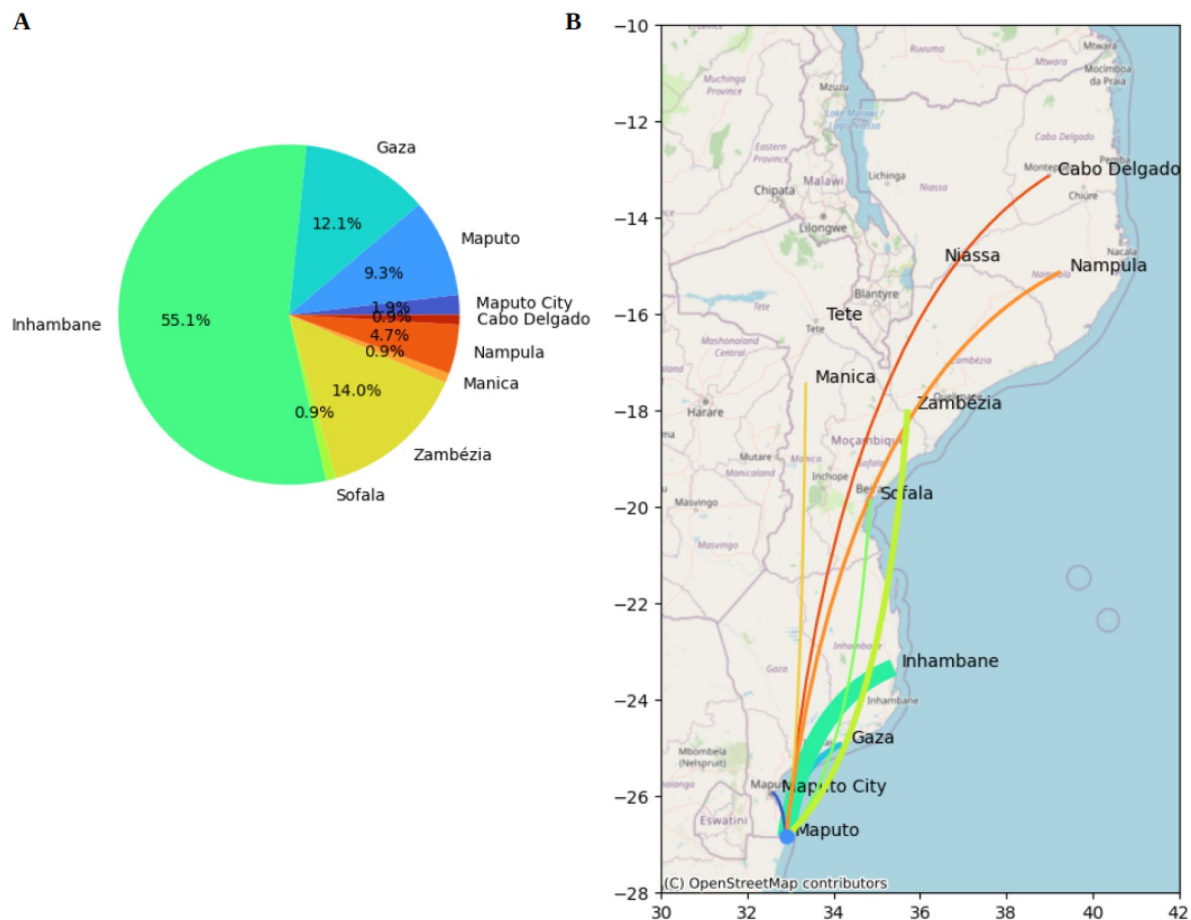


Figure 2

Statistics of travel reports.

A) Pie chart showing the distributions of the travel destination provinces in sampled individuals from Maputo province (Magude and Matutuine). Colours show the provinces from blue (south) to red (north). **B)** Spatial connectivity based on travel history. Line widths are proportional to the number of travels reported from Maputo province to their destination province, with the same colors as in A.

Category		Magude	Matutuine	p-value (Chi2)	p-value (T-test)
N		28	179	NA	NA
Dry season (%)		4 (14.3)	61 (34.1)	0.060	
Mean age (SD)		22.61 (14.9)	24.46 (17.6)		0.599
Female (%)		11 (39.3)	94 (52.5)	0.272	
Pregnant (%)		0 (0.0)	2 (3.0)	1.000	
Occupation, n (%)	Domestic	4 (14.3)	42 (23.5)	0.001	
	Student	9 (32.1)	17 (9.5)		
	Agriculture	3 (10.7)	22 (12.3)		
	Other	11 (39.3)	48 (26.8)		
	Unemployed	1 (3.6)	50 (27.9)		
Travelled (%)		3 (10.7)	104 (58.1)	<0.001	
Province travel destination, n	Gaza	2	11	0.357	
	Inhambane	1	58		
	Maputo	0	10		
	Zambézia	0	15		
	Manica	0	1		
	Maputo City	0	2		
	Sofala	0	1		
	Nampula	0	5		
	Cabo Delgado	0	1		

Table 1

Sample size and characteristics of the *P. falciparum* clinical cases in Magude and Matutuine with travel history data.

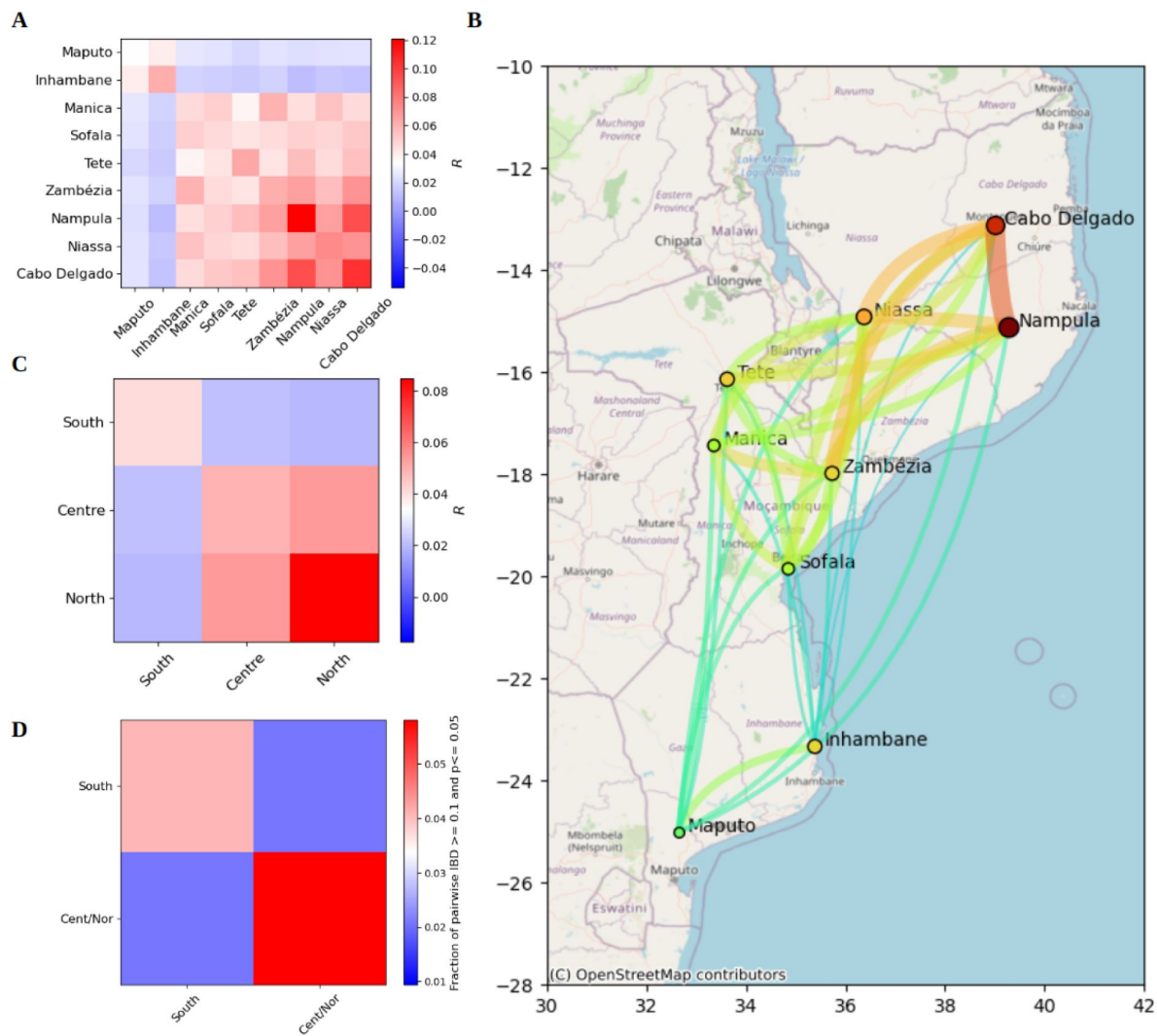


Figure 3
Genetic relatedness (identity-by-descent, IBD) of *P. falciparum* infections between provinces and regions in Mozambique.

A) Fraction (R) of IBD-related sample pairs (IBD > 0.1 with $p < 0.05$) within and between different provinces, represented in colours. B) Spatial genetic connectivity between provinces. Line widths and point sizes are proportional to R , and colours show the ranking in R values (from blue to red, using turbo colormap). C) R between and within different regions (south, centre and north), and combining samples from centre and north (D).

that trends do not depend on the threshold used) (Suppl. Fig. 4B). In the shortest distances between zero (the same household) and 10 km, the decrease of R with distance became significant again ($p < 0.001$) (Suppl. Fig. 4C).

***P. falciparum* importation rates**

A new Bayesian approach was used to classify clinical cases as imported and local. The method calculates an individual probability of being imported by combining epidemiological data (household-based mRDT positivity rates in children under 5 years per province from the demographic health survey 2022-2023⁴²), travel reports (date, duration and destination, interpreting infections as local if no travels were reported) and genetic IBD relatedness (R' between the sample and the parasite population in Maputo province, or the travel destination, see Methods) (Fig. 4). The importation probabilities obtained were in general close to 0 or 1, with only 1.9% (4/207) of them being between 0.25 and 0.75. Cases were classified as imported if their importation probability was higher than 50%. The fraction of imported cases within those who reported travel was 82.2% (88/107), corresponding to an importation rate of 42.5% (88/207) with respect to all cases. Among clinical cases from Magude and Matutuine with travel records, almost half (51.4%; 55/107) reported a trip to Inhambane province, representing 26.6% (55/207) of all studied cases from Magude and Matutuine (Table 2). Similar results were found when travel duration (or R') were not included in the estimation, being mRDT positivity rates the main driver of importation probabilities (Suppl. Importation analysis, Suppl. Fig. 5, Suppl. Table 3).

A statistically significant correlation was found between genetic relatedness and travel destinations. Cases reporting travel to Inhambane province presented a higher R' with the population from that province than cases reporting travels to other provinces or no travels at all ($p < 0.001$) (Suppl. Table 4). This correlation was not statistically significant for other travel destinations, possibly due to the lower sample sizes of travellers ($n \leq 15$).

Risk factor analysis of *P. falciparum* importation

Odds ratios (OR) were calculated using univariate and multivariate Firth's logistic regressions⁴³ to assess the risk factors associated with malaria importation ($n = 207$; Suppl. Table 5, Fig. 5). Pregnancy was excluded in the multivariate analysis due to the low number of pregnant women ($n = 2$). The residence district was strongly associated with importation rates in both univariate ($p < 0.001$) and multivariate ($p = 0.004$) analysis, with Magude district presenting a lower rate of importation (10.7%, 3/28) than Matutuine (47.5%, 85/179; OR = 6.6, 95%CI (2.3, 25.2), $p = 0.004$). Given that 96.6% (85/88) of the imported cases were from Matutuine district and only 3 were from Magude (Table 2), the statistics on imported cases, such as importation origin, mainly refer to Matutuine district.

In the univariate analysis, season, occupation and travel destination obtained statistically significant associations, with higher levels of importation in the dry season (OR = 2.0, 95%CI (1.1, 3.5), $p = 0.027$), lower importation rates for students (OR = 0.3, 95%CI (0.1, 0.8), $p = 0.013$), farmers (OR = 0.3, 95%CI (0.1, 0.8), $p = 0.019$) and other occupations (OR = 0.4, 95%CI (0.2, 0.8), $p = 0.009$) with respect to unemployed (or minors), and high importation rates ($> 80\%$, $p < 0.001$) for cases travelling to Gaza, Inhambane, Zambézia and Nampula (Suppl. Table 5, Fig. 5A) with respect to Maputo province. No associations were found for sex ($p = 0.075$), pregnancy ($p = 0.263$) or age ($p = 773$). In the multivariate analysis, only the residential district remained a significant factor ($p = 0.004$). No association was found between the rate of importation and parasite density ($p = 0.691$) or any estimation of polyclonality ($p > 0.052$ for any estimate). However, imported cases had a higher COI mean (2.82, 95%CI [1.0, 7.7]) and eCOI (1.77, 95%CI [1.0, 3.9]) than local cases (2.18, 95%CI (1.0, 5.1), $p = 0.005$ and 1.48, 95%CI (1.0, 3.3), $p = 0.008$, respectively) (Suppl. Table 5, Fig. 5A). Similar results were found when assessing association between these estimates and travel reporting (Suppl. Table 6, Fig. 5C).

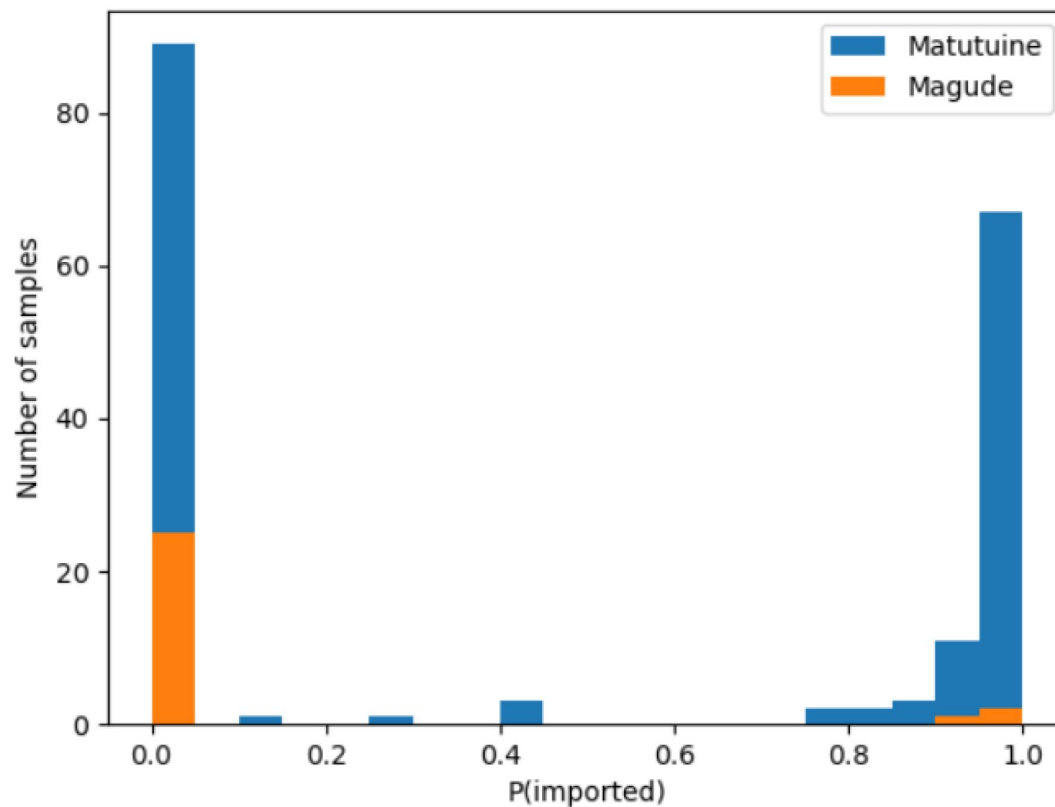


Figure 4

Distribution of importation probabilities by district.

Distribution of the individual probabilities of being imported for the studied clinical cases from Magude (orange) and Matutuine (blue) districts.

Importation province	Weighted number of cases	Imported cases	% cases with travel reports	% all reported	Imported to Magude	Imported to Matutuine
Cabo Delgado	1.0	1	0.9	0.5	0	1
Gaza	11.3	12	11.2	5.8	2	10
Inhambane	53.1	55	51.4	26.6	1	54
Manica	0.9	1	0.9	0.5	0	1
Maputo	2.7	1	0.9	0.5	0	1
Nampula	5.0	5	4.7	2.4	0	5
Sofala	1.0	1	0.9	0.5	0	1
Zambézia	11.9	12	11.2	5.8	0	12
Total	86.9	88	82.2	42.5	3	85

Table 2

Fraction of imported cases per province.

Weighted number of cases: the total number of cases (weighted by their probability) imported from each province (n=207). *Imported cases*: the total number of cases classified as imported if P(imported)>50% (n=207). *% cases with travel reports*: The fraction of imported cases from each province with respect to the total of cases reporting travels (n=107). *% all reported*: the total contribution of imported cases from each province with respect to all reported cases in the Maputo province (n=207). *Imported to Magude*: total of imported cases residing in Magude (n=28). *Imported to Matutuine*: total of imported cases residing in Matutuine (n=179).

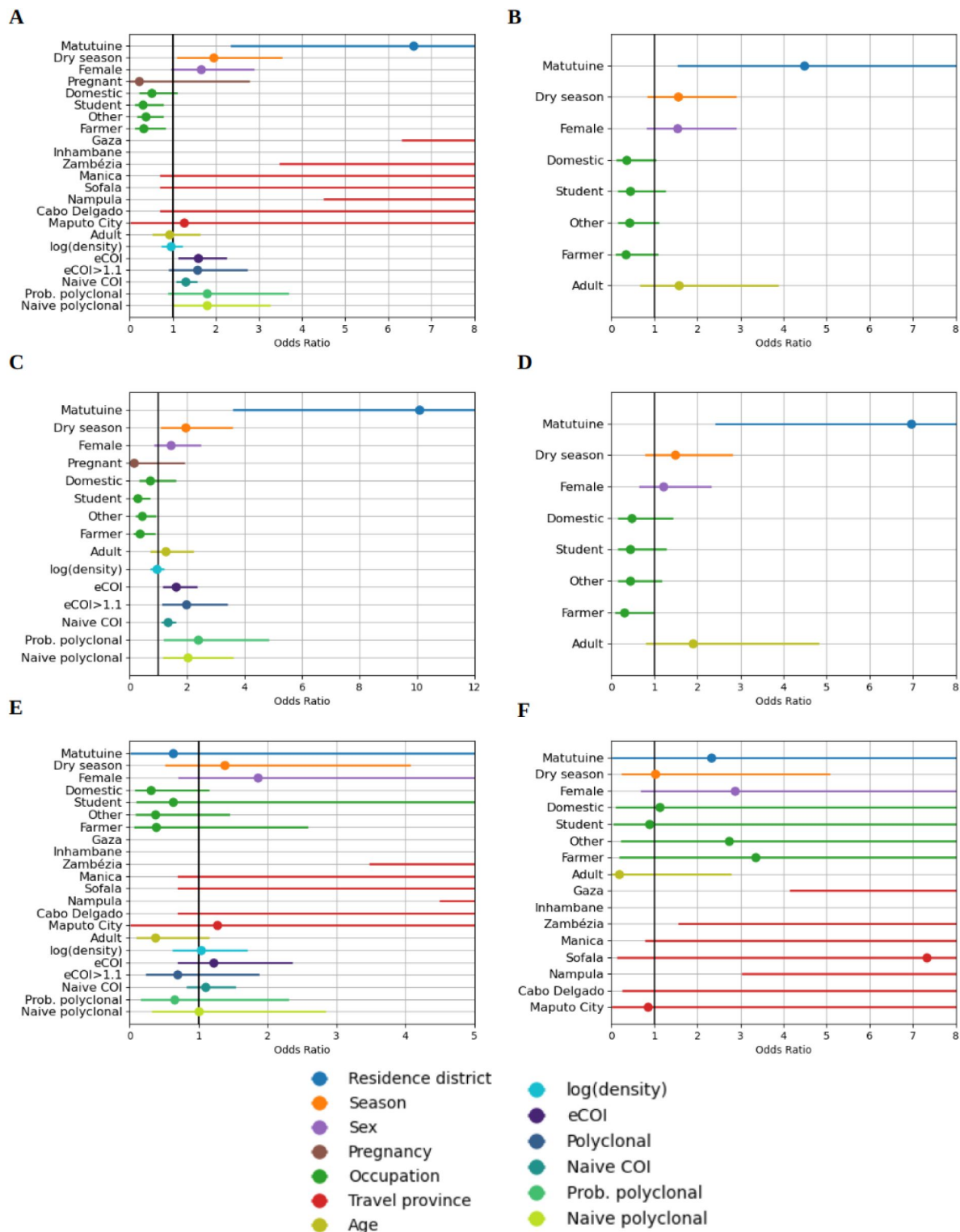


Figure 5

Odds ratio statistics of factors associated with importation and travel.

Odds ratio of importation (A-B), reporting travel (C-D) and importation for cases with travel reports (E-F) for different factors in univariate (A, C and E) and multivariate (B, D and F) models, for all *P. falciparum* clinical cases recruited in Magude and Matutuíne (n=207 for A-D, n=107 for E and F).

When restricting the analysis to cases reporting travel ($n=107$), the only factor that remained informative of importation was the province destination of the travel ($p<0.001$ for the province destinations of Gaza, Inhambane, Zambézia and Nampula), from both univariate and multivariate analysis (Suppl. Table 7, **Fig. 5E,F**). The probability of being imported was above 80% for all travels reported outside Maputo province and Maputo city, where the importation probability was below 10%. Neither parasitemia, COI nor polyclonality were found to be informative of importation.

Discussion

Knowledge of the sources of transmission, including imported cases, can inform the tailoring of effective targeted approaches for elimination^{40,44}. Here we have developed a novel Bayesian approach that allows the integration of genetic, travel history and epidemiological data to estimate probabilities of malaria importation. This new approach combines pairwise IBD estimates with travel reports to obtain a case-by-case probability of being imported. We found different importation rates between Magude (10.71%, 3/28) and Matutuine districts (47.5%, 85/179; OR=6.6, 95%CI (2.3, 25.2), $p=0.004$), in Maputo province, identifying Inhambane province as the main transmission source in Matutuine. COI and eCOI, which are within-host markers potentially informative of malaria burden in the population⁴⁵, were higher for imported cases, confirming our importation estimates from areas of higher malaria burden.

Our results show we can use parasite genomics to assess genetic population structure at the national level and at very-small scales (at the household level), but not at district level. We observed a high genetic relatedness between *P. falciparum* sample pairs within the centre-north of the country and a lower relatedness between south and centre-north. Pairwise genetic IBD-relatedness significantly decreased with the distance at large geographical scales, suggesting a strong isolation-by-distance at the country level, probably due to different parasite populations having different allele frequencies. No spatial correlation was found between samples at distances between 10 and 100 km, representing intra-province distances, indicating that *P. falciparum* genomics is heterogeneous at the provincial level. However, the spatial correlation was significant again at the smallest scales, including the distance of zero (the same household). This indicates that genetic IBD-relatedness is significantly higher for intra-household pairs, quickly decreasing to show no spatial pattern at small distances. The high relatedness of household members highlights the potential of genomics for very fine-scale transmission modelling approaches, such as transmission network modelling⁴⁶. The large-scale genetic structure suggests that *P. falciparum* genomics can be used to assess malaria importation across provinces, but the lack of structure at distances of 10-100 km might challenge the studies on importation within the province (across districts).

Cases notified during the dry season reported more travels (63.1%) and showed higher importation rates (53.8%) than those from the rainy season (46.5% and 37.2% respectively). However, there was no significant association with importation when conditioned to travel, implying that that importation and travel are highly correlated. This indicates that importation is higher precisely when transmission is lower. No association with importation was found for occupation, sex, age or pregnancy, suggesting that targeting specific subpopulations for malaria prevention or control might not be strategic.

A high fraction (51.7%) of *P. falciparum* clinical cases in Magude and Matutuine districts reported a travel in the last 28 days. Among them, a total of 82.2% of these cases were estimated as imported, representing 42.5% of all reported cases studied. Remarkable differences in importation rates were found between Magude (10.7%) and Matutuine (47.5%) districts, as well as in travel rates (10.7% in Magude and 58.1% in Inhambane). Inhambane was found to be the main source of importation in Matutuine, accounting for 63.5% (54/85) of all the imported cases from Matutuine.

Genetic relatedness confirmed these mobility patterns, with Matutuine showing higher genetic relatedness with Inhambane than Magude. Several factors can contribute to these different importation patterns. Matutuine district is close to Maputo City (a city with high mobility and importation levels), has better communication infrastructures and has some touristic places (e.g. Ponta do Ouro) of potential importation risk. In contrast, Magude district is an interior rural area, more isolated from big communication hubs, with less mobility due to work and with the National Kruger Park (South Africa) limiting movement in a significant fraction of its border. Further social studies would be required to identify those factors that increase the importation risk in southern Mozambique.

The results of this study have several practical implications for malaria elimination. First, improving the case classification through the obtained probabilities of importation conditional to the reported travel destinations allows for a better quantification of importation in elimination districts. Second, preventing importation, either by testing and treating travelers or by reducing malaria transmission in Inhambane, might be especially relevant to eliminate malaria in Matutuine district. These efforts would be more cost-efficient during the dry season when importation rates are higher. These interventions could potentially target around 60% of the infections in Matutuine, given that 60.9% of the studied cases are genetically more related to Inhambane than to Maputo. However, a better understanding of transmission networks is needed to quantify the real impact of importation on the overall transmission. Finally, targeting the whole population through vector control or mass drug administrations (MDAs) may be more appropriate in areas such as Magude where importation rates are low. In both districts, reactive strategies targeting remaining infections will be needed to interrupt transmission⁴⁷.

The study presents some limitations. First, except in Magude and Matutuine, *P. falciparum* isolates in the rest of the country were collected at selected health facilities and therefore may not be representative of the whole parasite population circulating in Mozambique, potentially biasing genetic relatedness with respect to Maputo province and potentially overestimating the genetic differentiation and underestimating importation. In particular, in Nampula, the province with the highest genetic relatedness, the sampling was conducted within 2 weeks in January and at only one health facility per district. Also, assuming that the samples from each province are representative of the travel destinations of the study cases might underestimate importation. However, the impact on our results must be small due to the high importation rates obtained. Second, only 25.6% (207/809) of the index cases were included in the importation analysis due to the low completeness of metadata. Third, our proxy of malaria transmission intensity was based on mRDT positivity rates in children under 5 years from a household-based survey, which is optimal for estimating malaria burden but not for transmission intensity. Also, importation probabilities relied on travel reports and were sensitive to their potential biases (e.g. unreported travels). Finally, seasonality of malaria transmission was not taken into account in the modelling of importation probabilities, which could increase the precision of the estimates. Future studies will use increased sample sizes and cluster sampling approaches, as well as refined modelling of malaria incidence, to address all these limitations.

To conclude, a new Bayesian model was used combining epidemiological, mobility and genetic data simultaneously to provide individual case classification, which potentially allow for identifying individual factors and specific populations for fine targeted approaches for elimination. Both mobility and genetic data were found to be informative of importation, highlighting the potential of malaria genomics to refine importation estimates. Very distinct importation and travel rates were found in two close districts with very similar malaria burden. The main sources of transmission were identified in these low-transmission areas that can inform decision-making strategies for malaria elimination.

Methods

Study design and sample collection

In Magde and Matutuine districts, all clinical cases who tested positive for malaria using HRP2-based rapid diagnostic tests (Bioline Malaria Ag *Pf.*, 05FK50 [Abbott], First Response® Malaria Antigen *P. falciparum* HRP2 [Premier Medical], AdvDx™ Malaria *Pf* Rapid Malaria Ag Detection Test [Advly]) during 2022 were invited to participate in the study if they were older than 6 months, resided in the area and had no symptoms of severe malaria. For all participants giving consent (by their adult representative for minors), forms were filled out to collect demographic and epidemiological information, including their travel times and destinations within the past 28 days. The RDTs were labeled with patient ID codes, placed in zip-lock plastic bags containing silica gel, and shipped to CISM where they were stored at -20°C until further processing.

A convenience sampling approach was conducted at selected health facilities in 9 provinces during the rainy season (January to May) of 2022^{38,40}; Maputo (including Maputo city; all ages >6 months); Inhambane, Manica, Zambézia, Sofala and Manica (children 2-10 years old); Nampula (children 3 months-5 years old); and Tete and Cabo Delgado (children 6 months-5 years old) (Suppl. Fig. 1). For all surveys, inclusion criteria were confirmed diagnosis of uncomplicated malaria by routine RDT, providing informed consent (participant or adult representative). For Tete and Cabo Delgado an additional criteria of parasite density >1000 parasites/μL was applied. Once consent was given, two to four 50μL dried blood spots (DBS) were prepared onto one or two filter papers through finger prick. DBS were identified with anonymous barcodes, air-dried, packed with silica gel and stored at 4°C until shipment to CISM, where sealed bags were kept at -20°C in the central laboratory until processing. The number of successfully sequenced samples per province ranged from 78 to 364, being the highest N for Inhambane province, the most frequent travel destination of the study cases (Suppl. Table 2). All samples were shipped to ISGlobal (Barcelona) for sample processing.

Genomic DNA extraction and quantification

Genomic DNA was extracted from RDT strips or from DBS samples using a Tween-Chelex based protocol³⁹ (see **Supplementary methods**). *P. falciparum* infection was confirmed in all DNA samples by qPCR targeting the 18S rRNA gene on an ABI PRISM 7500 HT Real-Time System (Applied Biosystems), as previously described⁴⁸. Parasite density was quantified by extrapolation to an external standard curve composed of six 1:10 dilutions of 3D7 cultured parasites in whole blood (range 1 to 100,000 parasites/μL; MRA-151, MR4, Bei Resources). DNA was stored at -20°C until sequencing.

Amplicon-based sequencing and sequence data analysis

Sequencing was performed using the MAD⁴HatTeR multiplex amplicon sequencing panel as previously described^{38,39} (**Supplementary methods**). FASTQ files were subjected to filtering, demultiplexing and allele inference using MAD⁴HatTeR Nextflow-based pipeline version 0.1.8 (<https://github.com/EPPIcenter/mad4hatter>)⁴¹. The 3D7 genome sequence was used as reference for alternative allele calling (https://github.com/EPPIcenter/mad4hatter/blob/main/resources/v4/ALL_refseq.fa). The resulting allele tables were subsequently filtered based on read counts and coverage across loci within a sample and across samples. Alleles with fewer reads than the maximum observed reads in any locus for negative controls were removed, along with alleles with <1% within-sample frequency.

For this analysis, the 165 loci of diversity from MAD⁴HaTeR D1 pool were used. Samples with less than 50 loci covered with a minimum of 100 reads were excluded. The exclusion criteria for loci was defined as those with less than 100 samples covering at least 100 reads.

Intra-host COI, eCOI and polyclonal probability were obtained using MOIRE v3.4.0⁴⁹ (<https://github.com/EPPIcenter/moire>), which uses a Monte Carlo Markov Chain (MCMC) approach taking into account intra-host relatedness and allele frequencies at the population level. Naive polyclonality was defined as COI>1, and polyclonality as eCOI>1.1. Genetic diversity was estimated as IBD using Dcifer 1.2.0⁵⁰ (<https://eppicenter.github.io/dcifer/>), which takes into account polyclonal infections and their intra-host relatedness to infer IBD across sample pairs. Dcifer provides an estimate of IBD as well as a p-value of rejecting IBD>0. In the analysis, sample pairs were defined as IBD-related if IBD>0.1 (higher thresholds implied lower statistics of related pairs) with p<0.05 (rejecting IBD=0), and the genetic relatedness, *R*, between two populations (or within a population) was defined as the fraction of IBD-related pairs between (or within) populations (equivalent trends were found for different IBD thresholds).

Statistical analysis

IBD analysis was conducted across provinces and across regions, defining regions as south (Maputo, Inhambane and Gaza), centre (Manica, Sofala, Tete and Zambézia) and north (Nampula, Niassa and Cabo Delgado). The statistical significance of *R* between (and within) populations was calculated as their difference with respect to the average *R* across the whole population. The p-value was calculated from bootstrap resampling with replacement, assuming no spatial differentiation.

Spatial structure of *R* (using IBD thresholds of 0.1, 0.15 and 0.2) was studied as a function of the pairwise geographical distance, using different distance ranges (100-1,400 km, 10-100 km and 0-10 km), with binning balancing spatial granularity with sample size. The statistical trends were obtained from a logistic regression of *R* versus distance, with a p-value of rejecting no dependence.

A new Bayesian approach was used to infer probabilities of importation combining epidemiological, genetic and mobility data. The approach models the probability of being infected (*I*) for *P. falciparum* in an area (*A*) given the *P. falciparum* genome (*G*) of the infection as:

$$P(I_A | G) = \frac{P(I_A)P(G | I_A)}{P(G)},$$

Where $P(I_A | G)$ is the probability of being infected in area *A* given a *P. falciparum* genome *G*, $P(G | I_A)$ is the probability of having a *P. falciparum* genome *G* if the infection occurred in area *A*, and $P(I_A)$ and $P(G)$ are the probabilities of being infected in area *A* and having *P. falciparum* genome *G* respectively. To estimate $P(I_A)$, it was assumed that the probability of an infection to have occurred in a province is proportional to a) the time spent in that province, and b) the transmission intensity of that province:

$$P(I_A) = K \times T_A \times PR_A,$$

Where T_A is the time spent in area *A*, PR_A is (a proxy of) the transmission intensity in *A* and *K* is an unknown constant that does not depend on the specific area, assuming that the differences in transmission intensity were captured in *PR* (**Suppl. Importation analysis**). $P(G | I_A)$ was estimated as *R'*, defined as the fraction of samples from *A* that are IBD-related to the sample studied. With the constraint that $P(I_A | G) + P(I_B | G) = 1$ if the case had only stayed in two areas recently (as the study cases), $P(I_A)$ becomes:

$$P(I_A) = \frac{T_A \times PR_A \times R'_A}{T_A \times PR_A \times R'_A + T_B \times PR_B \times R'_B}.$$

The probability of being imported was obtained defining A as the travel destination area and B as the local area (Maputo province). Assuming that the infection did not occur earlier than the past 28 days, a case was considered locally transmitted if it reported no travels in the last 28 days. The infection was assumed to take at least 7 days to become symptomatic, so T_A and T_B were obtained from the last 7 to 28 days. For missing travel durations, the average T of all available data was imputed, corresponding to 9.14 days. $PR_{A,B}$ were estimated from the mRDT positivity rates (PR_{RDT}) per province in children reported in the last Health Demographic Survey 2022-2023 from Mozambique.⁴² Since the PR_{RDT} in Maputo City was rounded as 0.0, a value of 0.04 was assumed, probably overestimating (with negligible impact) the infection probabilities in Maputo City. $R'_{A,B}$ was estimated as the fraction of samples from the area (province) that were IBD-related ($IBD > 0.1$, $p < 0.05$) with the sample studied. When estimating R' with Maputo province, one could consider excluding the samples from clinical cases reporting trips to avoid biases from imported cases. The impact of this choice was lower than 1% in the estimated imported rates, so all cases were included to estimate R' , a more conservative approach that was also more consistent with the other provinces where no travel reports were available. For missing R' estimates, such as in Gaza province, only T and PR factors were used to estimate probabilities of importation. Cases were classified as imported for importation probabilities higher than 50%, and as local otherwise. Importation rates were obtained from the fraction of imported cases over all classified cases. Similar rates were obtained using weighted sums of individual probabilities given the extreme values of probabilities obtained (Fig. 4, Table 2).

The correlation between travel reports and genetic relatedness was quantified from the fraction of cases reporting (or not) a travel in a given province and their fraction of cases that were more related to the origin and travel destination populations, conducting a chi square test for independence under the assumption that the probabilities of being more related to an area than to Maputo province were not correlated with travels reported.

Risk-factor analyses were conducted using first logistic regression to identify factors associated with importation and travel. The analysis included the following characteristics: district of residence (Magde or Matutuine), seasonality (defined as rainy for cases reported from January to May and dry for cases from June to December), sex, pregnancy, occupation, age (stratified as adults and minors) and province of travel destination. In order to assess the potential of molecular data to inform about imported cases, the same analysis was done including parasite density, eCOI and polyclonality. Since imported cases required a travel report by definition, the same factor analysis on importation was conducted restricted to those cases with travel reports (this time including travel destination in the multivariate analysis). Odds ratios and p-values were obtained from the first logistic regressions in a univariate analysis and also in a multivariate analysis using all factors without interactions (the limited sample sizes did not allow for exploring interactions between factors).

All over the analysis, statistical significance was defined as $p < 0.05$. All the analyses were performed using Python 3.9.16, Jupyter Lab 3.3.2 and R 4.2.3.

Ethical considerations

This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. Ethical approval was obtained from the Comité Nacional de Bioética para Saúde (CNBS) Mozambique, affiliated with the Ministério de Saúde (approval number 604/CNBS/21; 1st November 2023). The CNBS can be contacted at: Ministério de Saúde - 2º andar dto, Av. Eduardo Mondlane / Salvador Allende, PC 264, Maputo, Mozambique.

Data availability

Sequencing data is available at NCBI Sequence Read Archive (SRA) under BioProject accession code PRJNA1107381 (BioSamples SAMN45635392-SAMN4563593, SAMN41181143-SAMN41182039 and SAMN41180171-SAMN41180265).

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Additional information

Code availability

All code used in the analysis is open source under a GNU General Public License. The main open source repository of this analysis can be accessed here: https://github.com/MalPhyGen/malaria_relatedness_importation . The repository contains software requirements and installation instructions, with references to other public repositories used, and demos of the whole analyses conducted.

Author Contributions

A.P., A.C. drafted the manuscript and led the data analysis and interpretation of the results. C.dS., S.B., H.Mv., D.T., J.I., P.C., C.G.F., A.V.B. processed the samples and analysed the data. A.A.D. led the design of the sequencing panel and data processing. M.G.U. contributed to the bioinformatic pipeline and analysis for data processing and replicated the results of the analysis. E.R.V., A.M., B.G., A.A.D., M.M. contributed to the discussion and interpretation of results. C.G., S.M.E., F.S., P.A., B.C., A.M., B.R., E.R.V. participated in the study design. H.Mu., F.L., C.dS., L.N., W.S., N.C., A.C., G.M., N.N., M.T., J.M., L.F.S., K.U.B. coordinated the field work and data collection. All authors critically revised the manuscript, had full access to the data used in the study, and accepted the responsibility to submit it for publication.

Additional files

Supplementary Information [↗](#)

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Reviewer #1 (Public review):

Summary:

This study presents a new Bayesian approach to estimate importation probabilities of malaria, combining epidemiological data, travel history, and genetic data through pairwise IBD estimates. Importation is an important factor challenging malaria elimination, especially in low-transmission settings. This paper focuses on Magude and Matutuine, two districts in southern Mozambique with very low malaria transmission. The results show isolation-by-distance in Mozambique, with genetic relatedness decreasing with distances larger than 100 km, and no spatial correlation for distances between 10 and 100 km. But again, strong spatial correlation in distances smaller than 10 km. They report high genetic relatedness between Matutuine and Inhambane, higher than between Matutuine and Magude. Inhambane is the main source of importation in Matutuine, accounting for 63.5% of imported cases. Magude, on the other hand, shows smaller importation and travel rates than Matutuine, as it is a rural area with less mobility. Additionally, they report higher levels of importation and travel in the dry season, when transmission is lower. Also, no association with importation was found for occupation, sex, and other factors. These data have practical implications for public health strategies aiming for malaria elimination, for example, testing and treating travelers from Matutuine in the dry season.

Strengths:

The strength of this study lies in the combination of different sources of data - epidemiological, travel, and genetic data - to estimate importation probabilities, and the statistical analyses.

Weaknesses:

The authors recognize the limitations related to sample size and the biases of travel reports.

<https://doi.org/10.7554/eLife.107136.1.sa3>

Reviewer #2 (Public review):

Summary:

Based on a detailed dataset, the authors present a novel Bayesian approach to classify malaria cases as either imported or locally acquired.

Strengths:

The proposed Bayesian approach for case classification is simple, well justified, and allows the integration of parasite genomics, travel history, and epidemiological data. The work is well-written, very organized, and brings important contributions both to malaria control efforts in Mozambique and to the scientific community. Understanding the origin of cases is essential for designing more effective control measures and elimination strategies.

Weakness:

While the authors aim to classify cases as imported or locally acquired, the work lacks a quantification of the contribution of each case type to overall transmission.

The Bayesian rationale is sound and well justified; however, the formulation appears to present an inconsistency that is replicated in both the main text and the Supplementary Material.

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Reviewer #3 (Public review):

The authors present an important approach to identify imported *P. falciparum* malaria cases, combining genetic and epidemiological/travel data. This tool has the potential to be expanded to other contexts. The data was analyzed using convincing methods, including a novel statistical model; although some recognized limitations can be improved. This study will be of interest to researchers in public health and infectious diseases.

Strengths:

The study has several strengths, mainly the development of a novel Bayesian model that integrates genomic, epidemiological, and travel data to estimate importation probabilities. The results showed insights into malaria transmission dynamics, particularly identifying importation sources and differences in importation rates in Mozambique. Finally, the relevance of the findings is to suggest interventions focusing on the traveler population to help efforts for malaria elimination.

Weaknesses:

The study also has some limitations. The sample collection was not representative of some provinces, and not all samples had sufficient metadata for risk factor analysis, which can also be affected by travel recall bias. Additionally, the authors used a proxy for transmission intensity and assumed some conditions for the genetic variable when calculating the importation probability for specific scenarios. The weaknesses were assessed by the authors.

<https://doi.org/10.7554/eLife.107136.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

This study presents a new Bayesian approach to estimate importation probabilities of malaria, combining epidemiological data, travel history, and genetic data through pairwise IBD estimates. Importation is an important factor challenging malaria elimination, especially in low-transmission settings. This paper focuses on Magude and Matutuine, two districts in southern Mozambique with very low malaria transmission.

The results show isolation-by-distance in Mozambique, with genetic relatedness decreasing with distances larger than 100 km, and no spatial correlation for distances between 10 and 100 km. But again, strong spatial correlation in distances smaller than 10 km. They report high genetic relatedness between Matutuine and Inhambane, higher than between Matutuine and Magude. Inhambane is the main source of importation in Matutuine, accounting for 63.5% of imported cases. Magude, on the other hand, shows smaller importation and travel rates than Matutuine, as it is a rural area with less mobility. Additionally, they report higher levels of importation and travel in the dry season, when transmission is lower. Also, no association with importation was found for occupation, sex, and other factors. These data have practical implications for public health strategies aiming for malaria elimination, for example, testing and treating travelers from Matutuine in the dry season.

Strengths:

The strength of this study lies in the combination of different sources of data - epidemiological, travel, and genetic data - to estimate importation probabilities, and the statistical analyses.

Weaknesses:

The authors recognize the limitations related to sample size and the biases of travel reports.

Thank you for your review and consideration. As mentioned, we state in the manuscript the limitations related to sample sizes and travel reports. We aim to continue this study with new prospective data, aiming to address these limitations.

Reviewer #2 (Public review):

Summary:

Based on a detailed dataset, the authors present a novel Bayesian approach to classify malaria cases as either imported or locally acquired.

Strengths:

The proposed Bayesian approach for case classification is simple, well justified, and allows the integration of parasite genomics, travel history, and epidemiological data. The work is well-written, very organized, and brings important contributions both to malaria control efforts in Mozambique and to the scientific community. Understanding the origin of cases is essential for designing more effective control measures and elimination strategies.

Weakness:

While the authors aim to classify cases as imported or locally acquired, the work lacks a quantification of the contribution of each case type to overall transmission.

The Bayesian rationale is sound and well justified; however, the formulation appears to present an inconsistency that is replicated in both the main text and the Supplementary Material.

In fact, one of the questions that remains unanswered is the overall contribution of importation events to transmission in the areas. While the Bayesian classifier does not quantify this, our future analysis will focus on combining outbreak detection, genetic clustering and importation classification to quantify the contribution of imported cases to outbreak resurgence and to the overall transmission.

Thank you for pointing out the inconsistency in the final formula. In fact, the final formula corresponds to $P(I_A | G)$, instead of $P(I_A)$, so:

$$P(I_A | G) = \frac{T_A \times PR_A \times R'_A}{T_A \times PR_A \times R'_A + T_B \times PR_B \times R'_B}$$

instead of

$$P(I_A) = \frac{T_A \times PR_A \times R'_A}{T_A \times PR_A \times R'_A + T_B \times PR_B \times R'_B}$$

We will correct this error in a new version of the manuscript.

Reviewer #3 (Public review):

The authors present an important approach to identify imported P. falciparum malaria cases, combining genetic and epidemiological/travel data. This tool has the potential to be expanded to other contexts. The data was analyzed using convincing methods, including a novel statistical model; although some recognized limitations can be improved. This study will be of interest to researchers in public health and infectious diseases.

Strengths:

The study has several strengths, mainly the development of a novel Bayesian model that integrates genomic, epidemiological, and travel data to estimate importation probabilities. The results showed insights into malaria transmission dynamics, particularly identifying importation sources and differences in importation rates in Mozambique. Finally, the relevance of the findings is to suggest interventions focusing on the traveler population to help efforts for malaria elimination.

Weaknesses:

The study also has some limitations. The sample collection was not representative of some provinces, and not all samples had sufficient metadata for risk factor analysis, which can also be affected by travel recall bias. Additionally, the authors used a proxy for transmission intensity and assumed some conditions for the genetic variable when calculating the importation probability for specific scenarios. The weaknesses were assessed by the authors.

We acknowledge the limitations commented by the reviewer. We have the following plans to address the limitations. We will repeat the study for our data collected in 2023, which this time contains a good representation of all the provinces of Mozambique, and completeness of the metadata collection was ensured by implementing a new protocol in January 2023. Regarding the proxy for transmission intensity, we will refine the model by integrating monthly estimates of malaria incidence (previously calibrated to address testing and reporting rates) from the DHIS2 data, taking also into account the date of the reported cases in the analysis.

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